Vijay Kumaravelrajan

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Education

University of Southern California
B.S. Computer Science and B.S. Applied and Computational Math: GPA: 3.79 Spring 2026
Trustee Scholar - Highly Selective Full Tuition Merit Scholarship: 100 recipients selected out of all applicants
Relevant Coursework: AI, Data Structures, Mathematical Statistics, Probability Theory, Algorithms, Numerical Methods, Theory of Optimization

Experience

Jargon - Co-Founder November 2023 - Present
- Designing and deploying a Chrome extension for language learning, currently serving 100+ users
- Built a full-stack application using React, Supabase, and edge functions for scalable API infrastructure
- Implemented rigorous machine translation quality evaluation using industry benchmarks (FLORES+, WMT) and metrics (BLEURT, COMET)
- Developing an extensive LLM language translation comparison framework across multiple SOTA models, optimizing prompts and sampling methodologies for accuracy and serving efficiency
- Secured over $12,000 in competitive funding including IYA Ignite Grant to support development
- Conducting user experience initiatives through analytics implementation, user interviews, and iterative UI/backend improvements

USC VITERBI SCHOOL OF ENGINEERING

USC Center for AI in Society's Student Branch (CAIS++) - Project Lead December 2024 - Present
- Leading a team of 5 to research inference-time compute scaling for LLM language translation, achieving superior pass@k performance under a tighter inference budget compared to large models
- Implementing, analyzing various inference-time scaling strategies (best-of-n, tree search with PRMs, CoT with RL)

Previous Roles (Curriculum Lead) September 2024 - December 2024
- Mentored 3 underclassmen through practical ML projects and implementations through weekly curriculum meetings
- Taught comprehensive ML curriculum covering AI, neural networks, classical ML, NLP, CV, transformers, and generative modeling

Wireless Devices and Systems Lab - Directed Researcher September 2022 - August 2024
- Preprocessed 3D point cloud data using SLAM algorithms for Ray Tracer channel modeling in 6G wireless networks
- Employed heuristic-based and machine learning methods for edge detection in point clouds
- Refined Wireless Channel Modeling codebase, significantly improving runtime efficiency and readability.
- Designed a MATLAB GUI for Channel Modeling tooling to improve usability and streamline onboarding

Independent Projects

Empathy (empathy.zone) October 2024 - Present
- Leading development on AI-powered platform for finding meaningful connections through conversation analysis
- Implemented full-stack React application with Supabase, integrating multiple SOTA LLMs (Gemini, Claude, GPT-4o)
- Built custom voice agent using VAPI stack for natural conversation and artifact generation
- Successfully demonstrated platform to 300+ users across multiple hackathons, gathering feedback for refinement

LLM-Peer-Review (llm-peer-review.com)
- Created collaborative writing platform with Claude 3.5 Sonnet integration, attracting 200+ visitors
- Designed rich UI for seamless interaction with LLM-powered editing suggestions

Skills:

Python (Numpy, Matplotlib, PyTorch), MATLAB, JavaScript, SQL, C++, CSS, HTML, Java, React, JAX

Honors and Awards:

IYA Ignite Grants ($10000) | TroyLabs DEMO Second Prize ($2,000) | HackSC-X First Prize
High School GPA: 4.86 (The Quarry Lane School) | SAT: 1590